Roll No. | | | | | | | |

End Semester Examination June 2025

Name of the Course: CSE
Semester: IVth
Name of the Paper: Microprocessors
Paper Code: TCS 403

Time: 3 Hours
Maximum Marks: 100

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	CO1 & CO2
(a)	Explain the function of following pins: (i) HOLD (ii) BHE' (iii) MN/MX' (iv) DEN' (v) NMI	
(b)	Compare and contrast the architecture of the 8085 and 8086 microprocessors. Highlight key differences in terms of their data bus width, memory capacity and instruction sets.	
(c)	Draw the internal architecture of 8086. How does 8086 support pipelining?	
Q2	(20marks)	CO3
(a)	If a program uses each type of JMP, CALL and RET instruction exactly once in the 8085 microprocessor, calculate the total number of bytes consumed in memory. List all types of each instruction, mention their individual sizes, and show your calculations clearly.	
(b)	Write a program in 8086 which sort elements of an array in ascending order.	
(c)	Write a program in 8086 which determine subtraction of two 16 bit BCD number.	
Q3	(20marks)	CO2 &CO3
(a)	What are the errors in the following instructions of 8086? i) MUL 75 ii) MOV DS, 1234 iii) PUSH BH iv) ROR AL, 03 v) XCHG [DI], [BX]	
(b)	Draw 8086 flag register. Discuss the purpose and functioning of the control flags in the 8086 microprocessor.	
(c)	Explain relative addressing mode, string addressing mode, based indexed addressing mode and Input output addressing mode with two examples each.	
Q4	(20marks)	CO2 &CO4
(a)	Differentiate between a microprocessor and a microcontroller in terms of their architecture, functionalities, power consumption and applications.	
(b)	Draw and explain organization of RAM in 8051.	
(c)	Explain different types of interrupts of 8085. Discuss the benefits of using interrupts in real-time systems.	

Q5	(20marks)	
(a)	Write an 8085 program to toggle an LED connected to PC0 of Port C using the Bit Set/Reset (BSR) mode of the 8255. Include a delay between the ON and OFF states using a simple loop implemented with a register pair. Assume that the 8255 Control Register address is 83H.	CO5 & CO6
(b)	In an embedded system that uses the 8085 microprocessor, how would you interface an ADC (Analog-to-Digital Converter) to read analog signals from a sensor? Explain the process with a detailed example.	
(c)	Draw the block diagram of 8254. What are the applications of 8254 in real time system?	